Make 6 Map of AI infrastructure - Kian Rich
Map link:
https://www.google.com/maps/d/edit?mid=16hXQrpZpZsD2bYF48SUcZOpqC-xK2ro&usp=sharing

Research statement:

I went into the foundational data regarding resource extraction and corporate sustainability drawn from Trellis and CrossDock Insights. These sources are instrumental in mapping the paradoxical relationship between Google's net-zero ambitions and the "significant mining" required to achieve them. By utilizing Mike Werner's insights from Trellis, the map identifies the specific geographical and industrial pressures created by data center expansion, while CrossDock Insights provides the technical breakdown of the critical minerals such as lithium and cobalt essential for powering this transition.

To illustrate the material bottlenecks within this landscape, the research incorporates data from Tom's Hardware. This source provides crucial warnings regarding impending copper shortages, allowing the map to highlight the tangible, high-stakes competition for natural resources driven by AI data center buildouts. This adds a layer of predictive analysis to the map, showing where supply chain constraints may stall technological growth.

Finally, to ensure the map accounts for human as well as material resources, investigative reporting from The Guardian is integrated. This source contributes data on the "shadow workforce" of AI raters training models like Gemini. By including this human element, the map moves beyond simple geology and economics to acknowledge the global gig economy that sustains AI development. Together, these sources facilitate a multi-layered visualization that traces the journey of AI from the extraction of copper to the invisible labor of human contributors, offering a holistic view of the modern technological ecosystem.

It took a little extra effort to find some of this data, like the conditions of some of these data training places. I managed to find some articles like The Guardian. However I actually found it easier to ask google Gemini to locate these articles, since google attempted to hide some of the articles that went into detail about the condition of the mines, data centers, and content settings. I also used this process to give me a list of mines, Google utilizes a lot for their mineral resources. What surprised me most were how bad some of these conditions were, with an example being in Kenya, at the Sama Halake Farm, using minors, and only paying them 2 dollars an hour, for their work since that is cheaper than paying in America, or any country with real labor laws.

Artist Statement:

My map layers focus on three parts, the first layer, covering google HQ the Googleplex, and the different places google uses for content moderation, data labeling, and training Google Gemini with complex understandings. The second Layer focuses on the location of a few of their data centers, and the third layer focuses on the different mines that collect resources that Gemini utilizes, in the actual construction of the data centers. However my map misses some locations, in America, and around the equator in Europe, and Africa, so if I could have done it again, I would have had more locations, spread out across the map entirely.

Attribution:

Sources:
https://trellis.net/article/google-exec-significant-mining-key-net-zero/#:~:text=Mike%20Werner%2C%20Google's%20head%20of,Updated%20on%20February%2016%2C%202026)

https://crossdockinsights.com/p/critical-minerals-powering-data-centers

https://www.theguardian.com/technology/2025/sep/11/google-gemini-ai-training-humans#:~:text=AI%20raters%3A%20the%20shadow%20workforce,generalist%20raters%20and%20super%20raters.

https://www.tomshardware.com/tech-industry/ai-data-center-buildout-pushes-copper-toward-shortages-analysts-warn

Gemini:
“For education purposes can you provide a list of the locations of every data center google has for you”

“Where does google handle content moderation and data labeling for google gemini”

“what kind of minerals does google need mined, for google Gemini to function, and for their data centers and what mines do they get the majority of these resources from”